

Undecidability Does Not Kill Simulation: Provability, Execution, and the Missing Premise in Anti-Simulation Arguments

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Abstract

Suppose a process advances one state at a time under a specified rule, yet no algorithm can answer every global question about the process's indefinite future. Does the absence of that decision procedure, by itself, imply that the process cannot be generated computationally? It does not. The inference confuses at least three different objects: a generator of finite histories, a decider for global properties of those histories, and a semantic truth predicate for propositions about the system.

This paper reconstructs in its strongest form the argument in Faizal, Krauss, Shabir, and Marino's *Consequences of Undecidability in Physics on the Theory of Everything*. Their formal incompleteness claims are important, and the undecidability results they cite in many-body physics are real. The universal anti-simulation conclusion nevertheless requires an additional premise: physical adequacy must entail decision or certification of every relevant global truth. Gödel's, Tarski's, Chaitin's, and Turing's results do not supply that bridge. An effectively generated history can have undecidable global properties, as universal computation, undecidable dynamical systems, quantum thermalization, spectral-gap constructions, and computable-step renormalization-group flows demonstrate.

The strongest reply is that simulation should mean complete instantiation rather than finite prediction. That reply is coherent, but it sharpens rather than removes the missing-premise burden: the proposed nonalgorithmic semantic resource must be physically real, causally alter transitions, and resist every computable surrogate under the declared standard of adequacy. If those conditions are established, exact algorithmic simulation may indeed fail. They do not follow from incompleteness alone.

The argument is not strengthened by substituting randomness, Bell nonclassicality, computational complexity, inaccessible microhistory, or failed retrodiction for Turing noncomputability. A physical-oracle claim becomes testable only when it specifies an undecidable language, an effective encoding and preparation, a finite observer-time readout, a uniform reliability condition, precision and resource scaling, and repeatability assumptions. A result-by-result audit shows that the cited physics establishes several different obstructions: model-family classification limits, asymptotic decision limits, exact continuum noncomputability, and idealized observer-accessible hypercomputation. Exact continuum noncomputability is a genuine route by which a declared exact-state target can fail; Malament-Hogarth models are genuine theoretical evidence for observer-accessible hypercomputation under their spacetime and apparatus assumptions. Both are serious counterpressure against an unrestricted physical Church-Turing thesis. They show why the conclusion must be conditional on the representation of physical state and on what counts as simulation.

A fixed-commit Lean project by Jon Slotkin was independently reproduced. It machine-checks generation or theorem enumeration without decision under its declared definitions, but its `SimulationOf` predicate denotes an exact theorem-stream enumerator rather than a physical universe model. The formal result therefore verifies the generator-decider separation without settling the physical adequacy question. A causal-intervention and surrogate-equivalence framework is introduced to state what that missing physical bridge would require. The resulting conclusion is precise: undecidability alone does not rule out every declared simulation target; non-simulability follows only after a separately defended premise establishes a noncomputable physical transition or output, a usable hypercomputational protocol, or a semantic-omniscience requirement for simulation.

Keywords: undecidability; simulation hypothesis; physical Church-Turing thesis; hypercomputation; quantum gravity; Gödel incompleteness; causal intervention; probabilistic bisimulation; formal verification

1 The problem before the terminology

Consider a machine whose present state and update rule are known. Given the state now, the machine can compute its next state, then the next, and so on. Yet a different question may remain impossible to answer in advance: will this machine ever halt, enter a particular region, stabilize, thermalize, or converge to one of two long-term regimes?

These are not the same task. Advancing the system one step asks for a local operation. Settling what its unbounded history will eventually do asks for a global decision. A process can perform every next step without possessing a shortcut that classifies its entire future. The machine does not need to prove a theorem about that future before it moves.

That elementary separation is the center of the present dispute. Faizal, Krauss, Shabir, and Marino (FKSM) argue that Gödelian incompleteness, Tarskian undefinability, Chaitin-style information limits, and undecidable problems in mathematical physics prevent a wholly algorithmic Theory of Everything. They introduce a nonalgorithmic meta-theory and conclude that the universe cannot be a simulation [1]. The conclusion is stronger than the premises. The cited results constrain proof, semantic definition, compression, prediction, or general classification. To establish non-simulability, one must additionally connect one of those limits to the physical generation of the world.

The thesis developed here is therefore neither that the universe is a simulation nor that every mathematically possible physical model is Turing-computable. It is a diagnosis of the inference. If the universe contains a physically required noncomputable transition map or a usable hypercomputer, an algorithmic simulator may indeed fail. The burden is to establish that physical premise rather than infer it directly from incompleteness.

2 From an ordinary process to typed concepts

The distinction becomes exact once each word is assigned one job.

Execution applies a transition rule to produce another state. **Generation** produces a finite history or an output distribution under declared inputs and boundary conditions. **Prediction** computes a future state or statistic without necessarily reproducing every intermediate step. **Decision** returns a correct yes-or-no answer for every input in a defined problem class. **Proof** derives a sentence from axioms by formal rules. **Semantic truth** evaluates a sentence in a specified structure. **Explanation** supplies a scientific or metaphysical account and is not identical to any of

the preceding operations.

Other distinctions are equally necessary. **Computational complexity** measures resources needed by an algorithm; it does not by itself say that no algorithm exists. **Randomness** concerns outcome distributions or sequence properties; a random bit stream does not automatically decide a designated undecidable language. **Nonlocality** and **contextuality** constrain classes of physical models; they are not synonyms for non-Turing-computability. **Inaccessibility** and **irreversibility** constrain observation or reconstruction; they do not automatically constrain generation.

Let a discrete effective dynamical model have an encoded state $\mathbf{x}_t$, an admissible input or boundary condition $\mathbf{u}_t$, and a declared stochastic input $\mathbf{eta}_t$. Its next-state relation is

$$x_{t+1} = F(x_t, u_t, \eta_t). \quad (1)$$

When F , its inputs, and the stochastic interface are effectively specified, a finite-history generator can iterate Equation (1):

$$H_F(x_0, t) = (x_0, F(x_0), \dots, F^t(x_0)), \quad (2)$$

A global predicate asks a different question:

$$Q_F(x_0) = \begin{cases} 1, & \text{if the complete history has property } P, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

There need not be a total decider D_Q that returns $Q_F(\mathbf{x}_0)$ for every allowed input. The absence of D_Q does not erase F or any finite value of H_F .

3 The strongest form of the FKSM argument

A useful critique must first reconstruct the target in the form most favorable to it.

FKSM model a prospective quantum-gravity theory as a formal system FQG with a first-order language, a finite or at least recursively enumerable set of axioms, and effective inference rules [1]. The system is assumed to be consistent and expressive enough to encode arithmetic. Its theorem set is recursively enumerable. Under the standard hypotheses, Gödelian incompleteness follows: there are sentences that the system does not settle [2]. Tarski's theorem prevents an internal truth predicate satisfying the full material adequacy schema [3]. Chaitin's results provide related system-dependent information limits [4].

FKSM then identify the semantically true sentences of FQG with sentences true in the standard natural numbers and suggest that Gödel sentences correspond to empirically meaningful physical facts, including specific black-hole microstates [1]. Because the effective theory does not derive every such truth, they append an external truth predicate T , a non-recursively-enumerable set of truth axioms, and non-effective inference rules. Their meta-theory is intended to certify facts inaccessible to the algorithmic theory. Undecidable results concerning thermalization, spectral gaps, phase diagrams, tensor networks, and renormalization-group flows are presented as physical manifestations of the same need. Lucas-Penrose reasoning and objective-collapse proposals are then offered as possible expressions of nonalgorithmic understanding. Finally, because any simulation is taken to be algorithmic, FKSM conclude that no simulation can instantiate the complete meta-theoretic structure of the universe [1].

In compressed form, the inference is:

1. an adequate fundamental theory must be an effectively axiomatized formal system;
2. such a system is incomplete and lacks its own full truth predicate;
3. some unprovable arithmetical truths correspond to physical facts;
4. complete physical adequacy requires semantic certification of those facts;
5. physical evolution therefore requires a nonalgorithmic truth resource;
6. an algorithmic simulation cannot supply that resource;
7. therefore the universe cannot be simulated.

Steps 1 and 2 establish a genuine formal boundary. Steps 3 through 5 are the physical bridge. Step 6 adds a strong definition of simulation. The conclusion depends on those added premises, not on incompleteness alone.

4 The missing implication

The bare inference can be written as

$$\neg \exists D_Q \forall x [D_Q(x) = Q_F(x)] \implies \neg \exists H_F \forall x, t [H_F(x, t) = (x, F(x), \dots, F^t(x))]. \quad (4)$$

4.1 A finite-generation countermodel

Proposition 1 (finite generation does not entail global decision). There exists a computable transition map F , a computable input encoding $\mathbf{e}$, and a decidable state test $\mathbf{h}$ such that the state $F^t(\mathbf{e}(\mathbf{n}))$ and every finite history through time $\mathbf{t}$ are computable for every pair $(\mathbf{n}, \mathbf{t})$, while the property

$$Q_F(n) = 1 \iff \exists t \mathbf{h}(F^t(e(n))) = 1 \quad (5)$$

is undecidable.

Proof. Let F be the one-step transition function of a universal Turing machine, let $\mathbf{e}(\mathbf{n})$ encode a machine-input pair as an initial configuration, and let $\mathbf{h}$ return one exactly on halting configurations. A finite iteration of a computable transition map is computable, so $F^t(\mathbf{e}(\mathbf{n}))$ and its prefix are obtained by performing exactly $\mathbf{t}$ transitions. If Q_F were decidable, it would decide whether the encoded computation ever reaches a halting configuration. That is the halting problem, which is undecidable [5]. Therefore a total finite-step generator can coexist with an undecidable global property of what it generates. $\square$

This is not a verbal analogy and it does not prove that the universe is a Turing machine. A countermodel need not resemble quantum gravity to refute a universal logical entailment: it is enough that the antecedent can hold while the consequent fails. Proposition 1 therefore defeats only the inference from undecidability *alone*. It does not defeat the stronger proposition that our universe contains a separately established noncomputable transition, output, or constitutive resource. Any anti-simulation conclusion must supply that physical premise or otherwise rule out the generator-decider separation for its declared target.

Conway's Game of Life supplies a familiar visual illustration of the same separation: a local update rule can generate each new configuration while questions about the unbounded evolution inherit undecidability from universal computation. The broad use of Game of Life for this provability-versus-execution reply appeared in Redden's November 2025 preprint before the January 2026

article [6]. Proposition 1 uses the universal-machine form because it states the logical countermodel directly rather than relying on the illustration.

4.2 Adequacy is a typed profile, not an ordinal score

Let

$$K = \{\text{exec}, \text{op}, \text{dist}, \text{exact}, \text{dec}, \text{sem}\}$$

index six different success contracts. For each k , let $\mathbf{Gamma}_k$ specify the inputs, interventions, observables, representation, horizon, tolerance, and decision or certification burden appropriate to that target. The adequacy profile is

$$\mathbf{a}(S, W; \Gamma) = (A_k(S, W; \Gamma_k))_{k \in K} \in \{0, 1\}^K, \quad (6)$$

where $\mathbf{A}_k=1$ means that S passes the declared contract for target k . This definition deliberately does not assume that the six coordinates are linearly ordered. A finite generator can satisfy $\mathbf{A}_{\text{exec}}$ while failing $\mathbf{A}_{\text{dec}}$, as Proposition 1 proves. A decision oracle can answer a selected asymptotic question without generating the world's states. Exact state-at-time reproduction does not automatically supply a theorem prover for every semantic truth. If a study wants cumulative adequacy, it must define those conjunctions explicitly.

To derive non-simulability from the undecidability of a global property $\mathbf{Q}_W$, one needs a bridge such as

$$\forall S [\text{Algorithmic}(S) \wedge A_k(S, W; \Gamma_k) \implies \text{Decides}(S, Q_W)]. \quad (7)$$

If Equation (7) were true and $\mathbf{Q}_W$ were undecidable, no algorithmic simulator adequate for target k could exist. But Equation (7) is an adequacy premise, not a consequence of Gödel, Tarski, Chaitin, or Turing. A finite-horizon simulator can reproduce a history without deciding its infinite asymptotic properties. A distributionally adequate simulator can sample outcomes without explaining or certifying every individual result. A simulator can also be embedded, interactive, approximate, or open to boundary conditions. Which target or conjunction of targets counts is a scientific and philosophical choice that must be stated.

5 The strongest reply: instantiation rather than decision

FKSM can answer that the relevant target is not finite forecasting at all. On their strongest reading, the world contains a semantic or meta-theoretic resource that is constitutive of what the world is. A machine that reproduces selected outputs while omitting that resource would be a useful emulator, not a complete instantiation. If the external truth predicate T is genuinely part of physical reality, then demanding its preservation is not automatically an artificial request for an internal theorem prover.

That position deserves to be granted in full. It then acquires four empirical and mathematical obligations. First, T must be identified as more than an external description assigned by a theorist. Second, there must be a mapping from its nonalgorithmic contents to physical state transitions or observable probabilities. Third, the theory must identify either a replaceable physical interface whose supplied signal changes a downstream result or a noninterventional necessity relation that makes the

resource indispensable to the mechanism. Fourth, no computable surrogate may reproduce those consequences under the chosen adequacy predicate A_k .

If all four obligations are met, exact ontological or semantic simulation may fail. The operative premise would then be that physically realized evolution depends essentially on a noncomputable resource. Incompleteness can help characterize that resource, but it does not establish its physical existence or causal role. If T makes no difference to transitions, measurements, structural dependence, or mechanism-preserving comparisons, omitting it does not defeat finite, operational, or distributional adequacy. If it does make a difference, the anti-simulation argument succeeds through a noncomputable-dynamics or indispensable-resource premise rather than directly through Gödel’s theorem.

The strongest opponent is entitled to demand an exact-instantiation conjunction rather than accept any one coordinate of the adequacy profile. A suitable contract can require exact represented states, preservation of the operative transition structure, intervention-sensitive consequences, realization of every causally indispensable nonalgorithmic resource, and failure of every computable surrogate under an unbounded effective discriminator family. In compact form,

$$A_{\text{inst}} = A_{\text{exact}} \wedge A_{\text{mech}} \wedge A_{\text{int}} \wedge A_{\text{resource}} \wedge A_{\text{no-surrogate}}.$$

If a nonrecursive physical resource is shown to satisfy this contract, exact algorithmic instantiation is excluded. The disputed step is not the conditional conclusion. It is whether formal incompleteness, a semantic completion, or an undecidable physical classification already establishes every conjunct.

5.1 A typed truth-to-transition bridge

Let X_t be the physically operative state, U_t an admissible intervention or boundary input, R_t the declared stochastic input, q_t an effectively generated query index, and Θ the proposed nonalgorithmic semantic resource. Let Z_t be the physical interface variable alleged to couple a value derived from $\Theta(q_t)$ into the dynamics. The strongest FKSM mechanism would have to take a form such as

$$q_t = Q(X_t, U_t), \quad Z_t = C_{\Theta}(X_t, U_t, q_t), \quad X_{t+1} = G(X_t, U_t, R_t, Z_t). \quad (8)$$

Equation (8) does more than append the word *truth* to an ordinary evolution law. It names the query, the claimed coupling C_{Θ} , the physical interface variable, and the transition that consumes it. The model must then establish that C_{Θ} actually obtains the relevant nonalgorithmic value rather than merely being named after it. The remaining causal question is whether the final argument of G makes a difference.

Let $Y_{t+1:t+h} = H(X_{t+1:t+h})$ denote the declared observable consequences. Borrowing the distinction between observation and intervention from causal modeling [21], define the counterfactual effect of the proposed truth input in context $c = (x, u, q)$ by

$$\Delta_Z(c) = d_{TV}(P(Y_{t+1:t+h} \mid \text{do}(Z = z_1), c), P(Y_{t+1:t+h} \mid \text{do}(Z = z_0), c)). \quad (9)$$

The do operation in Equation (9) is a signal-replacement intervention on the proposed physical interface, not a proposal to make an arithmetical falsehood true. It asks whether blocking or replacing the value presented to the transition while holding the modeled context fixed changes downstream behavior. If $\Delta_Z(c) = 0$ for every admissible context, the proposed interface is causally idle in the model. If no signal, coupling, state variable, or transition argument corresponding

to Z can be specified, causal consumption has not yet been identified. A constitutive-identity view may reject intervention as the relevant test; it must then supply a necessity or mechanism-level no-surrogate argument instead. If effects occur only for a finite set of queries, a computable surrogate can hard-code those answers. If effects occur over an unbounded set but no effective preparation or finite readout exposes them, the model may differ ontologically without providing an observer-usable hypercomputer.

5.2 Surrogate equivalence is indexed by tests

Let I be a declared family of preparations and interventions, O a declared observable map, τ a finite horizon, and $\mathbf{M}$ the declared simulator model class—for example, uniform classical probabilistic machines, quantum channels with effective descriptions, or another precisely specified class. A surrogate S in $\mathbf{M}$ and target world model W are operationally equivalent to tolerance ϵ when

$$S \equiv_{I,O,\tau,\epsilon}^{\mathbf{M}} W \iff \sup_{i \in I} d_{\text{TV}}(P_S(O_{\leq \tau} \mid i), P_W(O_{\leq \tau} \mid i)) \leq \epsilon. \quad (10)$$

Equation (10) separates four questions that are often collapsed. Passive trace equivalence compares unperturbed outputs. Interventional equivalence compares outputs under the declared tests I . Mechanism equivalence additionally requires a state relation that preserves transitions; probabilistic bisimulation and testing theory make this distinction mathematically explicit [22]. Semantic identity adds the still stronger requirement that corresponding internal states carry the same interpretation or truth resource. A surrogate can pass the first two targets without being mechanism-identical, and it can be mechanism-corresponding under an abstraction without containing FKSM’s external predicate as a separately named object. Distributional adequacy therefore cannot be silently substituted for identity of an individual history or mechanism. Changing $\mathbf{M}$ also changes the no-surrogate burden; a theorem about uniform classical machines does not automatically exclude quantum, analogue, nonuniform, or stronger mechanism-preserving candidates.

Two no-surrogate results now become transparent. In the exact case, if effective preparation and finite readout make $Y(n) = \text{chi_L}(n)$ deterministically for every input of a nonrecursive language L , any algorithmic system exactly equivalent under those tests would compute chi_L , a contradiction. In the bounded-error case, the same conclusion holds within the declared classical probabilistic-machine model when the success gap is known and uniform, stopping and precision are effective, and the surrogate error is smaller than the gap [17, 20]. A finite test set cannot establish this conclusion: its answers can be hard-coded. Decisive exclusion therefore requires an unbounded, effective discriminator family together with effective preparation and readout. Removing any of those conditions reopens a loophole: nonuniform advice, a hidden noncomputable calibration, unbounded precision, non-effective stopping, or an intervention family too weak to expose the difference.

What incompleteness establishes - and what non-simulability additionally requires

The argument succeeds only when the formal limit is connected to a causally indispensable physical mechanism.

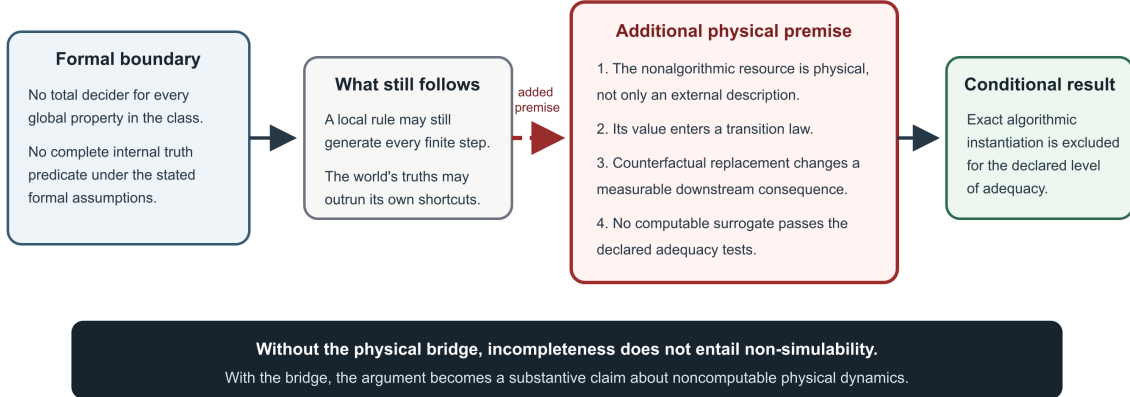


Figure 1. The missing physical bridge. Formal incompleteness blocks universal decision or internal semantic completion under the stated assumptions. Exact algorithmic non-simulability follows only after a nonalgorithmic resource is shown to be physical, transition-relevant, causally discriminable, and irreducible to a computable surrogate under the declared adequacy tests.

6 Undecidable physics often preserves the generator-decider split

The many-body and dynamical results invoked by FKSM make the distinction more concrete, not less.

Moore constructed dynamical systems whose long-term questions inherit undecidability from universal computation [7]. The result shows that an explicit dynamical description can coexist with no general decision procedure for a property of its unbounded behavior.

Cubitt, Pérez-García, and Wolf proved that the spectral-gap problem is undecidable for a constructed family of translationally invariant two-dimensional quantum spin systems in the thermodynamic limit [8]. The theorem is a profound limit on general classification. It does not say that a finite lattice has no Hamiltonian, that its local interaction terms are uncomputable, or that no finite-time evolution can be approximated.

Shiraishi and Matsumoto encoded a halting problem into a family of one-dimensional many-body systems and proved undecidability of a promised long-time thermalization question [9]. Their own limitations section is decisive: the result does not exclude proving thermalization or its absence for specific systems, and it does not diminish the meaning of finite-size numerical simulation. The theorem blocks a general ultimate criterion over the constructed family.

Watson, Onorati, and Cubitt provide the cleanest case. They construct an explicit renormalization-group map for an undecidable many-body family. Each renormalization step is computable; the map preserves the relevant phase information and converges to one of two fixed points. Yet which fixed point is reached is uncomputable in general [10]. The process can be iterated while its ultimate classification has no universal algorithmic shortcut. This is precisely the type separation that Equation (4) denies.

These results do not prove that the universe is computational. They prove that “a global physical question is undecidable” cannot be promoted, without another premise, into “the corresponding

physical update cannot run.”

6.1 Each physics result blocks a different operation

The phrase *undecidability in physics* is too coarse to carry an anti-simulation conclusion. The relevant results can be classified by the operation they block and the operations they leave intact.

Result	Encoded target	What the theorem blocks	What remains available	Missing operational bridge
Universal-machine countermodel [5]	Eventual halting	A total classifier for every encoded machine-input pair	Every finite transition and finite configuration	None is needed; this is the logical countermodel to Equation (4)
Moore dynamical systems [7]	Long-term reachability or asymptotic behavior	A general decision procedure for the encoded global property	An explicit dynamical rule and finite iteration	A uniform finite experiment that converts the asymptotic property into reliable oracle output
Spectral-gap family [8]	Gapped versus gapless thermodynamic-limit Hamiltonian	A general classifier over the constructed Hamiltonian family	Finite descriptions of local interactions and finite-volume calculations	Effective preparation of arbitrary encoded thermodynamic-limit instances and finite reliable readout
Thermalization family [9]	Promised long-time thermalization behavior	A universal long-time classifier over the constructed one-dimensional family	Specific-system proofs and finite-size numerical simulation	Uniform preparation and finite readout of the encoded asymptotic answer
Phase-diagram family [23]	Which of two phases occurs across a continuous Hamiltonian parameter	A general computation of the constructed phase diagram, including a positive-measure undecidable set	An explicit one-parameter Hamiltonian family and a phase-distinguishing order parameter	Uniformly preparing the required ground-state/thermodynamic regime for arbitrary encoded inputs without importing the answer

Result	Encoded target	What the theorem blocks	What remains available	Missing operational bridge
Renormalization flow [10]	Eventual fixed point or phase reached by the flow	A general algorithm for the ultimate classification	Every individual renormalization step and the explicit map	Finite termination or readout that decides which asymptotic basin is reached
Pour-El-Richards wave equation [11]	An exact solution value in a computable-analysis representation	Computability preservation of the exact continuum solution operator	The written wave equation and computable initial data under the paper’s representation	Finite-resource preparation, robust finite-precision measurement, and a uniform undecidable-language encoding
Malament-Hogarth proposals [15, 16]	Halting of an encoded computation	Ordinary Turing computability under the idealized spacetime and apparatus assumptions	A finite-observer readout after the computer’s unbounded proper-time history	Physical existence, stability, survivability, signal integrity, finite resources, and quantum/backreaction control

The table exposes a progression. The first six rows are theorem-level limits on classification, asymptotic behavior, or exact representation. The last row is qualitatively different because it already proposes a finite-observer protocol. Pour-El-Richards lies between them: it establishes exact noncomputable evolution, but not yet a robust laboratory oracle.

6.2 The phase-diagram result is the strongest many-body challenge

Bausch, Cubitt, and Watson construct a continuous one-parameter family of translationally invariant two-dimensional Hamiltonians whose phase diagram is uncomputable [23]. The two phases can be distinguished by an order parameter, and the paper states that the observable can be restricted to a single lattice site. The locality of that observable is a real strength: the result cannot be dismissed by saying that only an inaccessible global measurement could distinguish the phases.

The theorem still does not make the answer freely available to a finite experimenter. To use the family as an oracle, one must effectively encode an arbitrary input in the parameter, prepare the appropriate ground-state or thermodynamic-limit regime without already solving the hard problem, measure the order parameter with a known uniform reliability gap, and finish in finite observer time. If those steps can be supplied, the construction moves toward Equation (13). If they cannot, the system may instantiate one phase or the other while no general procedure decides which phase

was prepared. Physical possession of a property is not identical to algorithmic certification of that property.

6.3 None of the cited results makes the external truth predicate causal

The many-body results show that finite model descriptions can encode undecidable physical classifications. They do not show that the system queries FKSM’s external predicate T , that T changes a transition as in Equation (8), or that changing its supplied value changes a downstream distribution as in Equation (9). The results may motivate a search for noncomputable physical dynamics or operational oracles. They do not, without that extra construction, demonstrate the particular meta-theoretic mechanism proposed by FKSM.

7 Arithmetic truth is not yet a measured physical fact

FKSM define semantic truth using the standard model of arithmetic:

$$\text{True}(FQG) = \{\varphi \in L_{QG} : \mathbb{N} \models \varphi\}. \quad (11)$$

Equation (11) is a legitimate mathematical definition. It supports two readings that must remain separate. On the meta-theoretic-completion reading, the external truth set supplies truths unavailable to the effective formal theory; this bears directly on semantic omniscience. On the transition-level reading, values from that set are physically consumed by the world; this stronger claim requires a coupling such as Equation (8). The definition alone does not identify truth in $\mathbb{N}$ with a black-hole microstate, a measurement record, or a quantum-gravity transition. That identification needs a map from encoded sentences to physical preparations and observables.

For a physical system to operationalize an arithmetic truth oracle, one would need, at minimum, an effective preparation map E , a measurable readout M , and a reliability rule such that

$$M(E(\ulcorner \varphi \urcorner)) = 1 \iff \mathbb{N} \models \varphi \quad (12)$$

for the relevant class of sentences. If Equation (12) worked uniformly for an undecidable truth set, it would already specify a hypercomputational resource. Naming a black-hole microstate as an example does not construct E , identify M , prove the biconditional, or show how a finite observer extracts the answer.

The same issue applies to the external truth predicate. A non-r.e. set can be postulated mathematically. For it to explain a physical transition, one must state how its contents affect measurable dynamics, how a state queries it, and how different answers produce different outcomes. Without such a mechanism, the truth predicate labels the missing information rather than explaining how nature uses it.

8 What “simulation” must specify

The word *simulation* hides several targets:

1. **Finite execution:** generate a finite state prefix under specified rules.
2. **Operational approximation:** reproduce selected finite observables within a declared error and time horizon.

3. **Distributional adequacy:** reproduce measurement distributions under specified interventions.
4. **Exact state-at-time or trajectory adequacy:** return each requested represented state exactly; treat delivery of a completed infinite trajectory as a separate stronger demand.
5. **Global decision adequacy:** decide selected asymptotic properties for every input in a class.
6. **Semantic omniscience:** decide every truth about the represented world.

	k	Adequacy target	Required success	Direct relevance of the cited limits
	1	Finite execution	Generate a finite state prefix under the declared update rule and inputs	An undecidable asymptotic property does not by itself block finite iteration
	2	Operational approximation	Match named finite observables through horizon τ within tolerance ϵ	Exact continuum noncomputability may remain outside the declared finite tolerance
	3	Distributional adequacy	Match outcome distributions under declared preparations and interventions	Sampling does not require deciding or explaining every individual outcome
	4	Exact state-at-time or trajectory adequacy	Return each requested represented state exactly; a completed infinite object is a separate stronger demand	Pour-El-Richards-type noncomputable solution maps can block this target
	5	Global decision adequacy	Decide a named asymptotic property for every input in the class	Turing, spectral-gap, thermalization, phase-diagram, and RG results can block this target in their model families
	6	Semantic omniscience	Decide or certify every truth about the represented world	Gödelian incompleteness and Tarskian undefinability directly constrain this target under their hypotheses

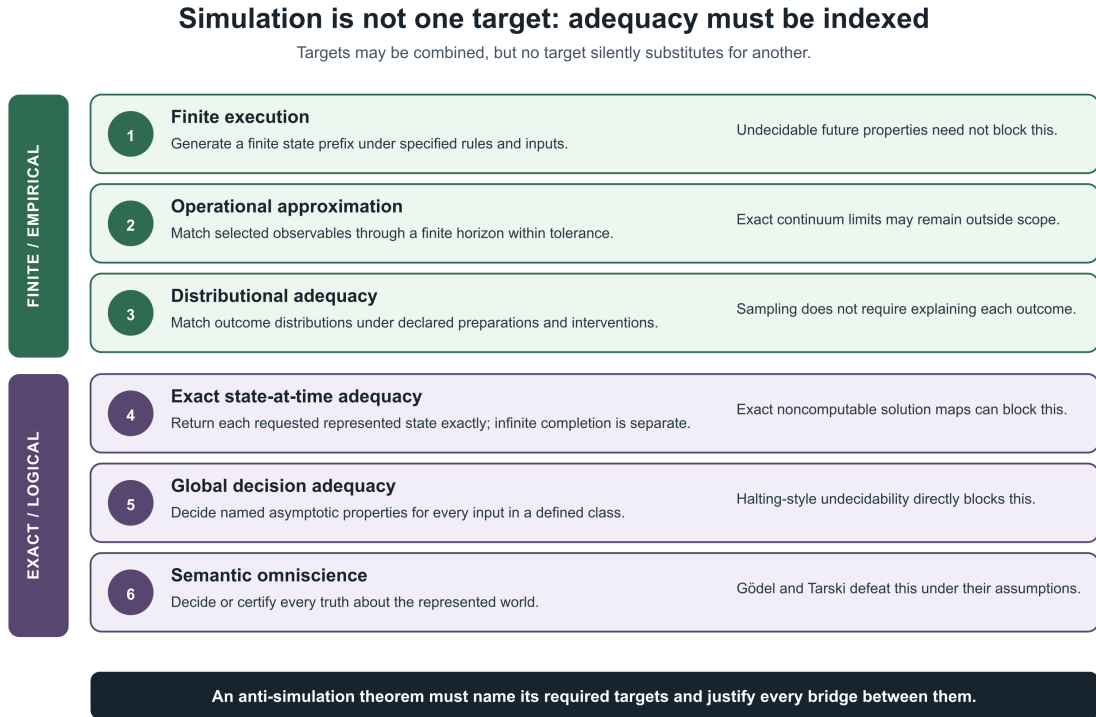


Figure 2. Six indexed targets of simulation adequacy. Targets 1-3 concern finite or empirical reproduction. Targets 4-6 concern exact state-at-time reproduction, global decision, or semantic certification. The indices label target classes; they do not by themselves create an implication ordering.

These targets are not automatically one linear hierarchy. A study may define a cumulative adequacy contract, but then the nesting relations must be stated rather than inferred from the index. Exact trajectory reproduction does not by itself supply a global decision procedure; a global decision oracle does not by itself generate a trajectory; semantic certification is yet another burden. Undecidability therefore bears differently on each target. A halting-style theorem directly blocks Target 5 for the relevant class and, in a sufficiently expressive effective theory, Target 6. It need not block Targets 1 through 3. Exact noncomputable solution maps can block Target 4 under the selected representation. A claim that the universe is not simulable is incomplete until it states which target or combination is required and why.

This also resolves the phrase *ontological sufficiency*. If it means reproduction of the world's operative states and observable consequences, the relevant question concerns transition and output maps. If it means internal certification of every true proposition, semantic omniscience has been added to the definition. Incompleteness can defeat that stronger target, but the result is then conditional on a requirement not normally imposed on a physical process. The universe does not appear to prove every theorem about itself before changing state.

9 The strongest counterpressure: exact continuum noncomputability

The generator-decider distinction cannot be inflated into a universal claim that every physically written evolution equation has a computable exact solution.

Pour-El and Richards constructed computable initial data for the three-dimensional wave equation whose unique solution takes a noncomputable value at a computable spacetime point under their computable-analysis framework [11]. This is a genuine mathematical case in which computability is not preserved by an exact continuum evolution operator.

The lesson is not to dismiss noncomputable dynamics but to type the claim. A proposed physical consequence depends on:

- the representation and norm under which the initial data are computable;
- the regularity and energy properties of those data;
- whether the preparation can be specified and realized with finite resources;
- whether the noncomputable quantity is a finite-precision observable;
- whether noise and perturbation preserve the claimed effect; and
- what accuracy and horizon the simulator is required to achieve.

Pour-El and Richards therefore defeat a careless universal premise—“effective equations always have computable exact solutions.” They do not automatically provide a laboratory device that decides the halting set. Their result supports one possible added premise for anti-simulation reasoning: a physically required exact transition or output map may be noncomputable in the relevant representation. Establishing that premise for our universe would be scientifically consequential.

This conditional posture matches the physical Church-Turing literature. Gandy derived computability for mechanisms under substantive conditions such as bounded information density and bounded propagation [12]. Deutsch’s universal quantum computer changes the complexity landscape without computing nonrecursive functions in the model [13]. Arrighi and Dowek give a quantum version of the Gandy strategy and emphasize that physical computability follows from explicit physical assumptions, not logic alone [14].

10 The strongest counterpressure: relativistic hypercomputation

Malament-Hogarth spacetimes require a second correction to casual dismissals of “exotic” oracles. In the idealized geometry, one worldline has infinite proper duration while an observer reaches an event in finite proper time that can receive a signal from that entire worldline. Hogarth explicitly uses this structure to propose decisions of Turing-unsolvable problems, including the halting problem [15]. Etesi and N emeti develop related black-hole computer designs and address several objections within classical general relativity [16].

These are genuine mathematical hypercomputation proposals. They are not merely stories in which information is hidden somewhere. Their remaining burden is physical:

- Does a spacetime with the required global structure exist or arise?
- Can a computer survive and perform the unbounded sequence of steps?
- Can the observer receive a distinguishable signal without destructive blueshift or instability?
- Do mass inflation, backreaction, quantum effects, finite entropy, or Planck-scale cutoffs invalidate the construction?
- Can the absence of a signal be interpreted reliably as non-halting at the observer event?

The correct conclusion is neither that general relativity trivially supplies an oracle nor that hypercomputation is meaningless. It is that an idealized decision protocol exists in the model, while its physical realizability remains the premise that would connect it to our universe.

10.1 Zeno, Cauchy-horizon, and Hawking-radiation claims are different routes

The January 2026 article grouped Malament-Hogarth spacetimes, Zeno processes, Cauchy horizons, and proposals that answers might be encoded in Hawking radiation [19]. That grouping correctly demanded a fixed standard of evidence, but it understated one item: a Malament-Hogarth model is not merely a machine running for an arbitrarily long time. It is designed so that a finite-proper-time observer receives a halting signal after the computer worldline has unbounded proper duration [15]. Etesi and Némethi’s black-hole scenarios investigate a concrete general-relativistic realization involving the relevant horizon structure [16].

The other motifs must be evaluated separately. A Zeno label describes an accelerated infinite-step idealization unless a particular spacetime and apparatus model is supplied. A claim that information is present in Hawking radiation is a claim about physical encoding or recoverability; it is not yet a procedure that maps arbitrary encoded inputs to correct decisions for an undecidable language. Even complete information preservation would not, by itself, provide an effective decoder, a finite stopping rule, or a uniform reliability gap.

The article’s deeper distinction survives the correction. Three standards must not slide into one another:

1. **Model-level possibility:** a mathematical model contains a finite-observer decision protocol.
2. **Physical realizability:** the required geometry, apparatus, signal, resources, and stability conditions obtain in our universe.
3. **Intuitive mystery:** a process is inaccessible, exotic, information rich, or difficult to reconstruct.

Malament-Hogarth models meet the first standard under their assumptions. Whether they meet the second remains open. The third standard, by itself, has no computability-theoretic force.

11 Randomness, nonclassicality, complexity, and hidden information

Several replies to the January 2026 article changed the vocabulary without supplying the missing bridge. The distinctions can be stated cleanly.

11.1 Randomness

A genuinely random infinite sequence may be noncomputable with probability one. That does not turn its individual bits into correct answers for a chosen undecidable language. An oracle is not merely an unpredictable source; it provides structured decision power. Ordinary probabilistic computation with effective transition probabilities does not obtain unrestricted nonrecursive decision power merely by sampling [17]. A noncomputable probability parameter or noncomputable correlation could change this conclusion, but then that parameter is the added resource that must be physically justified.

11.2 Bell nonclassicality and contextuality

Bell-type results constrain local hidden-variable completions. Tsirelson-type bounds constrain achievable correlations. Neither theorem states that no Turing-equivalent substrate can sample the operational distributions of quantum theory. A computational substrate need not be a local deterministic hidden-variable theory.

11.3 Complexity

An exact classical simulation of a generic quantum system may require prohibitive resources. That is a profound practical and complexity-theoretic limitation. It is distinct from proving that no Turing machine computes the required function. Quantum advantage can transform feasibility without crossing the recursive/nonrecursive boundary [13, 14].

11.4 Inaccessibility and retrodiction

Black-hole horizons, erased microhistories, irreversible processes, and hidden internal degrees of freedom can prevent an observer from reconstructing how an outcome arose. Failure of access is not itself a decision procedure and does not show that the process was not generated. Information existing somewhere is also not enough: the question is whether a finite observer can extract a reliable answer to a specified undecidable problem.

11.5 Explanation

A simulator can reproduce operational statistics without deciding among metaphysical interpretations of those statistics. Demanding an explanation of why one outcome occurred adds a different adequacy target. Generation, prediction, decision, and explanation must not be exchanged mid-argument.

12 The operational oracle criterion

The January 2026 article’s most useful discriminator is the oracle question, now expressed as a protocol rather than a rhetorical challenge.

Let L be a specified undecidable language. A physical system is evidence for usable hypercomputation only if there exists an effective input encoding and preparation **Prep**, a declared physical evolution E , a finite observer readout **Read**, and a uniform success condition. As a minimum bounded-error form, require some $\delta > 0$ such that

$$\text{Prep}(n) \downarrow, \quad \tau_E(n) < \infty, \quad \Pr\left[\text{Read}\left(E_{\tau_E(n)}(\text{Prep}(n))\right) = \chi_L(n)\right] \geq \frac{1}{2} + \delta \quad \text{for every } n, \quad (13)$$

with declared precision, energy, space, runtime, and termination assumptions. If repeated trials are used to amplify reliability, the preparation and readout must be independently repeatable at the required scale rather than reuse a hidden noncomputable calibration. Beggs, Costa, and Tucker show why the apparatus model matters: measurement precision and runtime can change the computational class attributed to a physical oracle [20]. Stronger exact or arbitrarily amplifiable criteria may be appropriate depending on the model.

When does a physical system become evidence for hypercomputation?

The test requires a complete input-to-answer protocol, not unpredictability alone.

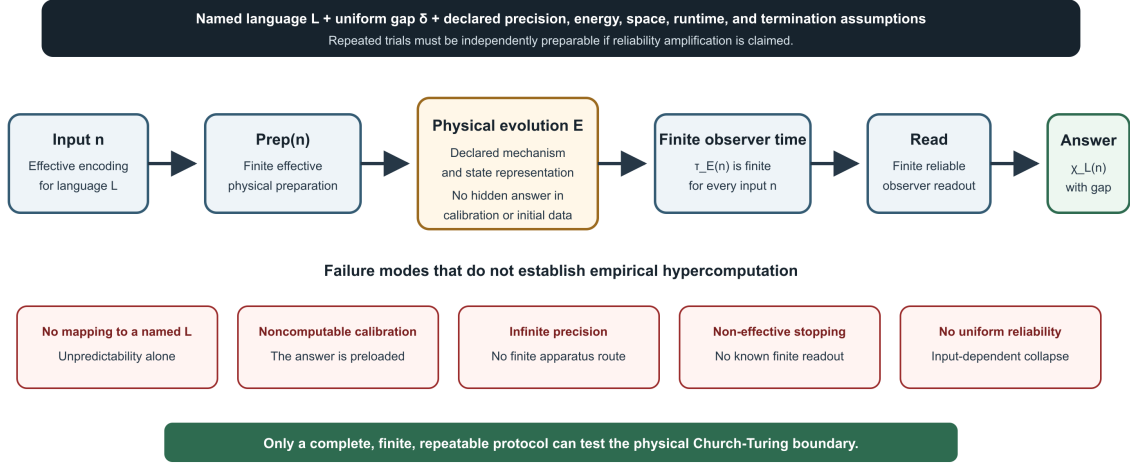


Figure 3. Operational oracle test. A named undecidable language must be connected to effective preparation, declared physical evolution, finite observer time, reliable readout, and a uniform success condition. Unpredictability, hidden noncomputable calibration, infinite precision, non-effective stopping, or input-dependent reliability do not satisfy the test.

This criterion separates five cases:

1. unpredictable output with no mapping to L ;
2. a mapping that assumes a noncomputable constant in the preparation;
3. a mathematical infinite-time or infinite-precision protocol;
4. an idealized finite-observer hypercomputer whose physical assumptions remain disputed; and
5. a realizable, reliable protocol that would refute the ordinary physical Church-Turing thesis.

Only the fifth establishes empirical hypercomputation. The fourth is important theoretical evidence. The first three do not by themselves show that nature decides an undecidable language for an observer. This observer-readout criterion is not imposed on every ontological claim. A noncomputable transition could be physically indispensable without yielding a usable oracle to embedded experimenters; that possibility belongs to the exact-dynamics and constitutive-resource routes, not to the empirical-hypercomputer route.

13 Machine-checked evidence and its boundary

Jon Slotkin’s `simulation-undecidability` repository formalizes a generation-versus-decision separation in Lean. This study fixed commit `ee340ee186e8589ef3782d719ca744a1675e5153`, reproduced the project using Lean 4.32.0 and Lake 5.0.0, and obtained 828 successful build jobs, 39 binding axiom-audit lines, exit code zero, and no admitted proofs in the audited targets [18]. The audit command traverses each listed declaration’s axiom closure and fails the build if it encounters `sorryAx`, a custom axiom, or anything outside `propext`, `Classical.choice`, and `Quot.sound`.

The formal result is useful, but its definitions must control its interpretation. In the repository:

- `ObservationStream` α is a time-indexed stream of finite words, but the central `SimulationOf` definition is stated directly over a process $P : \mathbb{N} \rightarrow \text{Option } S$;
- `Theory` provides an effectively coded sentence type, a computable negation map, and a recursively

enumerable provability predicate;

- **Theory.RepresentsInseparablePair** captures the recursion-theoretic interface used for the Robinson-arithmetic hypothesis; the concrete **DiagTheory** witness is not claimed to literally interpret arithmetic;
- **AlgorithmicToE** T is a total Boolean decider for $T.Prov$;
- **AdequateEnum** T P means that process P emits every theorem of T and only theorems of T ;
- **SimulationOf** T P means P is computable and satisfies that exact theorem enumeration condition;
- **PStar** T says that the existence of such a simulation entails an algorithmic provability decider, while **FksmInference** T is its contraposed form: no algorithmic provability decider entails no such simulation;
- **WorldTruth** is a bivalent predicate over a diagonal universal-machine construction.

Lean verifies that theorem enumeration can coexist with incompleteness, and that a computable history can induce an undecidable and non-r.e. semantic truth set. It also verifies a counterexample to the repository’s encoded inference from absence of a provability decider to absence of an adequate theorem enumerator. These results are theorems under the repository’s interfaces; the names *empirical adequacy* and *simulation* in comments do not enlarge the formal definitions.

What Lean does not verify is that theorem-stream enumeration constitutes physical simulation. **AdequateEnum** does not encode trajectories, measurement distributions, Hamiltonian evolution, continuum states, empirical tolerance, or intervention. The name **SimulationOf** is a modeling label at that boundary. The machine-checked result establishes a countermodel to a formal generator-decider implication; the mapping from that result to physical adequacy remains an independent argument. Machine checking cannot validate a prose-to-physics bridge that has not itself been formalized.

14 What would actually establish non-simulability?

The missing premise can be supplied in at least three scientifically distinct ways.

14.1 A noncomputable physical transition or output map

Identify a physically necessary map under an effective preparation and measurement representation, then prove that no Turing machine computes the required outputs to the declared accuracy. Exact continuum examples show that this is mathematically possible. A claim about our universe must add physical accessibility and robustness.

14.2 A usable hypercomputer

Demonstrate a finite observer protocol meeting Equation (13) for a named undecidable language. This would be direct evidence that physical evolution supplies more than Turing computation.

14.3 A defended semantic-omniscience definition

Require a simulator to decide every truth about the simulated world. Gödelian and Tarskian limits can defeat that target. The conclusion then concerns semantic omniscience, not ordinary finite execution or empirical simulation. The philosophical work is to justify why semantic certification is constitutive of physical instantiation.

These routes are not equivalent. The first concerns dynamics or outputs, the second concerns observer-accessible computational power, and the third concerns a definition of adequacy.

15 The theorem-level burden-of-proof result

The paper establishes a conditional structure rather than a positive simulation theorem.

1. Proposition 1 disproves the unqualified entailment from absence of a global decider to absence of a finite-step generator.
2. Equation (7) shows how non-simulability would follow for a declared target if adequacy for that target entailed decision of an undecidable property.
3. Equations (8) through (10) state how a proposed nonalgorithmic resource could instead become physically decisive through a coupling, downstream consequence, and no-surrogate result.
4. Equation (13) states the additional requirements for an observer-usable hypercomputer.
5. The exact-instantiation conjunction grants the strongest ontological objection: if a nonrecursive resource is physically indispensable and every computable surrogate fails the declared effective discriminator family, exact algorithmic instantiation is excluded.

None of these results establishes that our universe is simulated. The result is that the universal anti-simulation conclusion is underdetermined until one of the additional physical or definitional bridges is established.

16 Chronology and contribution

The broad proof-versus-execution objection did not first appear in the January 2026 article. Evan Redden’s November 2025 preprint *Provability vs. Execution* already argued that an incomplete formal description can still execute, used Conway’s Game of Life, and identified physical hypercomputation as the missing empirical burden [6]. That prior work should be credited.

The January 2026 SVGN article developed a wider diagnostic independently in public discussion [19]. Its additional contribution was to isolate the arithmetic-to-observable encoding problem, analyze FKSM’s external truth predicate as an operationally unspecified oracle, separate Turing noncomputability from randomness, nonlocality, classical inefficiency, horizon inaccessibility, and failed retrodiction, and formulate a concrete oracle question. The present paper turns that question into Equation (13), introduces an explicit indexed family of simulation-adequacy targets, incorporates exact continuum and relativistic counterpressure, and audits the later Lean formalization against its definitions.

This chronology strengthens the argument by separating inherited insight, Micah Blumberg’s distinct extensions, and later machine-checked support.

17 Discussion

Undecidability in physics is not superficial. It can be encoded into spectral gaps, long-time observables, phase diagrams, and renormalization flows. Exact solution operators can fail to preserve computability. General relativity admits idealized spacetime structures designed to expose a finite observer to an infinite computation. None of these results should be reduced to “mere ignorance.”

Their force depends on what has been proved noncomputable. A global classification theorem may leave every finite transition effective. A noncomputable exact continuum value may remain

inaccessible at finite precision. An idealized hypercomputer may possess genuine decision power in a mathematical spacetime while failing under physical stability constraints. Conversely, if any of these bridges is established operationally for our universe, the computational picture must change.

The strongest exact-instantiation objection remains live. A simulator that matches every finite observation while omitting a causally indispensable nonalgorithmic ingredient would not instantiate the same mechanism. The response is not to redefine that concern away, but to require its decisive bridge: identify the ingredient, show how it enters the transition law, demonstrate an intervention-sensitive consequence, and exclude computable surrogates at the declared level of adequacy. Without that bridge, semantic difference alone has not yet become physical difference. With it, the case against exact algorithmic instantiation would be substantive.

This is why the dispute cannot be resolved by the word *undecidable* alone. The scientific object must be typed: theorem set, semantic truth set, transition rule, exact solution operator, observable, probability distribution, asymptotic classifier, or finite decision protocol. Only then can the result bear on simulation.

The most defensible residue of FKSM is substantial. No finite effective formalism should be expected to prove, compress, predict, or classify every truth about an arithmetically expressive world. Some physical model families contain questions for which no universal decision procedure exists. Internal agents cannot expect an omniscient shortcut around every emergent computation. Those conclusions constrain a complete algorithmic *oracle about reality*. They do not, without an additional premise, constrain an algorithmic process from generating a reality whose truths outrun the process’s own capacity for certification.

18 Conclusion

A rule can generate a history without deciding every global property of that history. A formal theory can enumerate consequences without defining its complete semantic truth set. A stochastic process can generate operationally correct distributions without turning randomness into a halting oracle. A many-body map can have computable individual steps while its final regime remains undecidable.

Accordingly, incompleteness and undecidability do not by themselves establish that the universe cannot be simulated. The inference requires a further premise: adequate physical simulation must decide the relevant global truths, or the physical transition/output map must itself be noncomputable, or nature must provide a usable hypercomputational protocol. Each is a coherent research proposition. None follows automatically from Gödel, Tarski, Chaitin, or the existence of undecidable classification problems.

The correct scientific question is therefore not whether physics contains undecidability. It does. The question is where the noncomputability resides, what representation makes it physical, what finite observer can measure, and which standard of simulation is being tested. Once those variables are made explicit, the argument stops sliding between proof, truth, prediction, execution, and explanation. The possibility of non-simulability remains open, but it must be established through the missing physical bridge rather than declared as a corollary of incompleteness.

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